

Riley Beenders

FORWARD DEPLOYMENT | AI-ENABLED PRODUCTS | AUTOMATION

Bethlehem, PA | [\(484\) 523-4431](tel:(484)523-4431) | rileybeenders@gmail.com | www.rileybeenders.com | linkedin.com/in/riley-beenders

PROFESSIONAL SUMMARY

Product-minded R&D and engineering leader with 10+ years of multidisciplinary experience delivering working systems from problem definition through commercialization and support. Builds automation, remote operations, AI-assisted diagnostics, and internal business systems while partnering directly with users, vendors, and cross-functional stakeholders.

CORE EXPERTISE

AI & Automation: AI-assisted diagnostics; machine-vision integration; workflow automation; LabVIEW/Modbus; Python; local LLM tooling (LM Studio); Git/GitHub

Product Delivery: Problem definition; product development and commercialization; user workflows; troubleshooting; technical support; documentation; operator training

Enterprise Systems: ERP/CRM development; Kubernetes; Docker and containerized applications; Windows/Linux administration; network/VLAN configuration; risk analysis

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Own the mechanical, electrical, and software design of ICARUS-Lite from product development through commercialization, technical support, and after-sales service.
- Implemented PROTEOR Print with an external software partner, enabling remote operation, AI-assisted diagnostics, live video, and material, waste, and outcome metrics.
- Built an automated LabVIEW/Modbus QC workflow that collects networked sensor data, displays live tolerances, and generates traceable PDF reports for field support.
- Managed Microsoft Business Central ERP implementation for the Coplay site, aligning SKUs and operating workflows with Proteor's global locations.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

CTO | Lead Electro-Mechanical Engineer | Industrial Engineer | Engineering Technician

- Progressed through four roles to direct the product-development pipeline, internal IT infrastructure, and technology strategy while remaining hands-on across software, firmware, electrical, mechanical, and manufacturing systems.
- Designed and hosted a redundant Kubernetes-based ERP/CRM for orders, inventory, tasks, projects, and customer interactions, with dynamic BOM generation connected to purchasing.
- Designed and launched Gen3 POSEIDON, integrating industrial-servo motion, safety controls, firmware, and a customer-facing hybrid extrusion platform.
- Delivered custom automated systems from CAD and BOMs through sourcing, assembly, calibration, operator training, troubleshooting, and post-delivery support.

ADDITIONAL RELEVANT EXPERIENCE

Northampton Area School District - Computer/Network Technician | 2016 - 2019 | Automated 2,500+ device deployments with an Arduino-based setup tool and deployed 750+ wireless access points.

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and manufacture 3D-printing hardware accessories.

EDUCATION & PROFESSIONAL DEVELOPMENT

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Relevant development: Machine Learning, MATLAB, Simulink, and Simscape Onramps (2026); Lean Six Sigma Yellow Belt (2025); Penn State Certificate in Engineering Design and Tools (2023)